Industrial Vision

Lecture 11 - Estimating homography

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Where we are

L12 gave the forward model $\tilde{\mathbf{x}} \sim P\tilde{\mathbf{X}}$ with $P = K[\, R \mid \mathbf{t} \,]$. In general the result **depends on the depth** of the world point, so two views are not related by any single matrix: near things shift more than far things, and that difference is parallax.

There are exactly **two circumstances in which the depth cancels**, and in each of them one 3×3 matrix relates the two images completely.

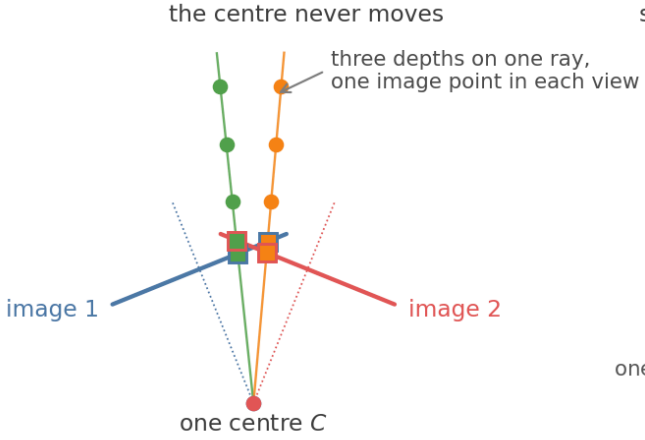
- The camera **rotates about its own centre**, so that $\mathbf{t} = \mathbf{0}$. No parallax can arise, whatever the scene.
- The scene is a **plane**, so that depth is not a free variable but a function of position on that plane.

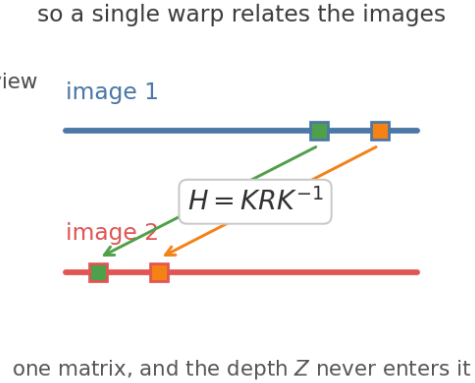
L10 built the hierarchy of plane transformations and put the **homography** $\tilde{\mathbf{x}}_t \sim H\tilde{\mathbf{x}}_s$ at the top of it, with nine entries, one free scale and **eight degrees of freedom**. What it did not say is which H a given camera produces. Today has both halves.

1. We **derive** H from the P of L12, for each of the two cases above, and see how a warp is applied to a grid of pixels.
2. We convert the relation "$\sim$" into linear equations by the **direct linear transform**, and show that four correspondences suffice.
3. We make the estimate robust to matches that are noisy and contain outliers, by **RANSAC**.
4. We solve the over-determined system by the **singular value decomposition**.
5. We ask **which error that solution actually minimizes**, and find it is not the one we want.
6. We stitch a **panorama**.

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First case: a camera that only turns





The first of the two special cases is a camera that **rotates about its own centre**, so that $\mathbf{t} = \mathbf{0}$. Placing the world origin at that centre, the two views of the same point $\tilde{\mathbf{X}}$ are

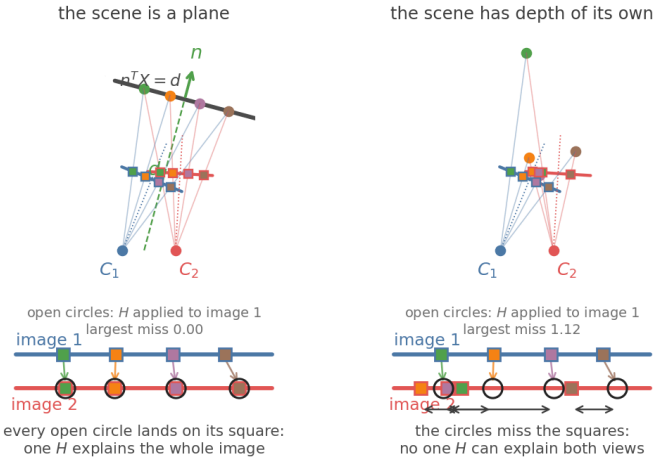
$$\tilde{\mathbf{x}}_1 = K[\,I \mid \mathbf{0}\,]\tilde{\mathbf{X}}, \qquad \tilde{\mathbf{x}}_2 = K[\,R \mid \mathbf{0}\,]\tilde{\mathbf{X}}$$

From the first equation $K^{-1}\tilde{\mathbf{x}}_1 \propto \mathbf{X}_{1:3}$, and substituting this into the second gives

$$\tilde{\mathbf{x}}_2 \propto K R K^{-1}\tilde{\mathbf{x}}_1 \quad \Rightarrow \quad \boxed{H = KRK^{-1}}$$

The scene depth Z **cancels**, so that one matrix maps every point irrespective of its depth.

Second case: a planar scene, and the general result



- Now let the camera **translate as well**, but let the scene lie on a **plane**, written in the frame of the first camera as $\mathbf{n}^\top \mathbf{X} = d$, with $\mathbf{n}$ the unit normal and $d > 0$ the perpendicular distance from the first camera centre to the plane. Putting the first camera at the origin: $\tilde{\mathbf{x}}_1 \sim K\mathbf{X}, \tilde{\mathbf{x}}_2 \sim K(R\mathbf{X} + \mathbf{t})$
- The whole derivation is one trick. Because $\mathbf{X}$ lies on the plane, $\mathbf{n}^\top \mathbf{X} = d$ and the number $\mathbf{n}^\top \mathbf{X}/d$ equals **one**, so $\mathbf{t}$ may be multiplied by it without changing anything. The translation thereby becomes a linear function of $\mathbf{X}$, and the terms collect:

$$\mathbf{t} = \mathbf{t} \cdot \frac{\mathbf{n}^\top \mathbf{X}}{d} \quad \Rightarrow \quad R\mathbf{X} + \mathbf{t} = \left(R + \tfrac{1}{d}\mathbf{t}\,\mathbf{n}^\top\right)\mathbf{X}$$

- Since $\mathbf{X} \propto K^{-1}\tilde{\mathbf{x}}_1$,

$$\boxed{H = K\left(R + \tfrac{1}{d}\mathbf{t}\,\mathbf{n}^\top\right)K^{-1}}$$

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From the relation $\sim$ to explicit equations

For a single correspondence $(x_s, y_s) \leftrightarrow (x_t, y_t)$ there exists some $\lambda \neq 0$ such that

$$\lambda \begin{bmatrix} x_t \\ y_t \\ 1 \end{bmatrix} = H \begin{bmatrix} x_s \\ y_s \\ 1 \end{bmatrix} \qquad \Longleftrightarrow \qquad \begin{aligned} \lambda x_t &= h_{11} x_s + h_{12} y_s + h_{13} \\ \lambda y_t &= h_{21} x_s + h_{22} y_s + h_{23} \\ \lambda &= h_{31} x_s + h_{32} y_s + h_{33} \end{aligned}$$

The quantity λ is a nuisance parameter and is **eliminated** by dividing the first two rows by the third:

$$x_t = \frac{h_{11} x_s + h_{12} y_s + h_{13}}{h_{31} x_s + h_{32} y_s + h_{33}} \qquad y_t = \frac{h_{21} x_s + h_{22} y_s + h_{23}}{h_{31} x_s + h_{32} y_s + h_{33}}$$

These expressions are nonlinear in (x_s, y_s), but **cross-multiplying** makes them linear in the unknowns h_{ij}:

$$\begin{aligned} h_{31} x_s x_t + h_{32} y_s x_t + h_{33} x_t &= h_{11} x_s + h_{12} y_s + h_{13} \\ h_{31} x_s y_t + h_{32} y_s y_t + h_{33} y_t &= h_{21} x_s + h_{22} y_s + h_{23} \end{aligned}$$

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The direct linear transform

We collect the unknowns into the vector $\mathbf{h} = (h_{11}, h_{12}, h_{13}, h_{21}, h_{22}, h_{23}, h_{31}, h_{32}, h_{33})^\top \in \mathbb{R}^9$. Each correspondence contributes **two rows**:

$$\begin{bmatrix} -x_s & -y_s & -1 & 0 & 0 & 0 & x_s x_t & y_s x_t & x_t \\ 0 & 0 & 0 & -x_s & -y_s & -1 & x_s y_t & y_s y_t & y_t \end{bmatrix} \mathbf{h} = \mathbf{0}$$

- There are eight unknowns, since the scale does not count, so **four correspondences** suffice and give $A\,\mathbf{h} = \mathbf{0}$ with $A \in \mathbb{R}^{8\times 9}$.
- A system of size 8×9 possesses a non-trivial null vector, and this vector is the homography. It is determined only up to scale, since $A(\alpha\mathbf{h}) = \mathbf{0}$ also holds, which is consistent with the eight degrees of freedom established earlier.
- This procedure of cross-multiplying, stacking and extracting the null space is the **direct linear transform**, or DLT.

Where do the four correspondences come from? They are obtained by running SIFT, from Lecture 7, on the two images and taking the matched keypoints . This is precisely where the difficulty begins.

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Matches are noisy, and some are incorrect

Two distinct problems arise.

Noise. The matches are correct but their coordinates are imprecise, typically by about a pixel.

The remedy is to use many correspondences together with least squares, so that the errors average out.

Outliers. Some matches are simply incorrect, arising when similar descriptors are computed at different scene points.

Least squares does **not** remedy outliers, since a single gross outlier can distort the entire fit. The exact four-point DLT is more vulnerable still, since one incorrect match renders H meaningless.

Solution: **RANSAC**

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RANSAC: hypothesis and verification

Given m correspondences $(\mathbf{x}_i \leftrightarrow \mathbf{x}'_i)$ and a tolerance t, the following steps are repeated.

1. **Select four** correspondences, the minimal sample, at random and independently of every previous iteration; solve $A\mathbf{h} = \mathbf{0}$ and obtain a candidate H.
2. **Apply H to all the remaining points**, computing $\lambda\,(x''_i,\ y''_i,\ 1)^\top = H\,\tilde{\mathbf{x}}_i$ and normalizing by the third element to return to pixel coordinates.
3. Compute the **transfer error** in pixels in the target image:

$$\varepsilon_i = \sqrt{(x'_i - x''_i)^2 + (y'_i - y''_i)^2}$$

   SIFT asserts that $\mathbf{x}_i$ corresponds to $\mathbf{x}'_i$, whereas the candidate H maps it to $\mathbf{x}''_i$, and a good H brings the two into agreement.
4. If $\varepsilon_i < t$, for example one pixel, then point i joins the **consensus set** of this candidate.
5. **Terminate** when the consensus set is sufficiently large, say 60 % of m. Otherwise repeat for a fixed number of iterations and retain the H with the **largest consensus**, which is not in general the last one computed. Nothing else is carried from one iteration to the next.
6. **Re-solve using all the inliers.** The four-point estimate should not be used as the final answer.

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The over-determined system $A\mathbf{h} = \mathbf{0},\ A \in \mathbb{R}^{m\times 9},\ m \gg 8$

In the presence of noise no vector $\mathbf{h}$ satisfies all the equations exactly, so we seek the best solution:

$$\min_{\mathbf{h}} \|A\mathbf{h}\|^2 \quad \text{subject to} \quad \|\mathbf{h}\| = 1$$

The constraint fixes the arbitrary scale. Without it the minimum would be attained at $\mathbf{h} = \mathbf{0}$.

Expanding the objective gives

$$\|A\mathbf{h}\|^2 = (A\mathbf{h})^\top (A\mathbf{h}) = \mathbf{h}^\top A^\top A\,\mathbf{h}$$

We now introduce the **singular value decomposition** of A:

$$A = U\,\Sigma\,V^\top, \qquad U^\top U = I_{m\times m}, \quad V^\top V = V V^\top = I_{9\times 9}, \quad \Sigma^\top \Sigma = \operatorname{diag}(\sigma_1^2, \dots, \sigma_9^2)$$

The singular values are ordered so that $\sigma_1 \geq \cdots \geq \sigma_9 \geq 0$, and the quantities σ_i^2 are the eigenvalues of $A^\top A$.

$$\mathbf{h}^\top A^\top A\,\mathbf{h} = \mathbf{h}^\top V \Sigma^\top \underbrace{U^\top U}_{I} \Sigma V^\top \mathbf{h} = \mathbf{h}^\top V\,(\Sigma^\top \Sigma)\,V^\top \mathbf{h}$$

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The minimizer is the last right singular vector

We substitute $\mathbf{h} = \mathbf{v}_9$, the last column of V, and evaluate the product.

- First, $V^\top \mathbf{v}_9 = (0, \dots, 0, 1)^\top$, since the rows of $V^\top$ are orthonormal and only the last row has a non-zero inner product with $\mathbf{v}_9$.
- Next, $(\Sigma^\top \Sigma)\,(0, \dots, 0, 1)^\top = (0, \dots, 0, \sigma_9^2)^\top$.
- Finally, $V\,(0, \dots, 0, \sigma_9^2)^\top = \sigma_9^2\,\mathbf{v}_9$, and $\mathbf{v}_9^\top(\sigma_9^2 \mathbf{v}_9) = \sigma_9^2$.

The choice $\mathbf{h} = \mathbf{v}_9$ therefore gives $\|A\mathbf{h}\|^2 = \sigma_9^2$, which is the **smallest** value attainable by any unit vector. This follows because writing an arbitrary $\mathbf{h}$ as $V\mathbf{y}$ with $\|\mathbf{y}\| = 1$ gives $\sum_i \sigma_i^2 y_i^2 \geq \sigma_9^2$.

$$\boxed{\hat{\mathbf{h}} = \mathbf{v}_9 = \text{last column of } V} \qquad \text{and the } 9 \times 1 \text{ vector is reshaped into a } 3 \times 3 \text{ matrix}$$

If the correspondences were exact then $\sigma_9 = 0$ and $\mathbf{v}_9$ is a true null vector, so that the noiseless case is recovered as a special case.

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Practical considerations

Normalize the coordinates. Pixel coordinates, of order 10^3, cause the columns of A to differ greatly in magnitude, which makes $A^\top A$ ill-conditioned. The standard remedy, due to Hartley, is to translate the points so that their centroid is at the origin, to scale them so that their mean distance from the origin is $\sqrt{2}$, to run the DLT, and then to undo the normalization:

$$H = T_t^{-1}\, \hat{H}\, T_s$$

This costs little and improves the estimate substantially.

Degeneracy. If three of the four points are **collinear** the system loses rank and H is not unique. Implementations of RANSAC reject such samples.

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An application: panorama stitching

Five overlapping images I_1 to I_5 are captured by rotating a hand-held camera. The procedure is as follows.

1. Choose the **middle** image, say I_3, as the reference. The motion between the two extreme images would be too large, whereas from the center it remains moderate.
2. Create a wide empty **canvas** around the frame of I_3.
3. Estimate H_{13}, H_{23}, H_{43} and H_{53} using SIFT, RANSAC and the DLT, each computed pairwise with respect to the reference.
4. Perform a **backward warp**, mapping each canvas pixel through the appropriate homography into the source image and reading the intensity there. This leaves no holes, unlike a forward mapping.
5. Handle **overlap** by combining the values a canvas pixel receives from several images, taking the **median** rather than the mean, since the median also suppresses ghosting.

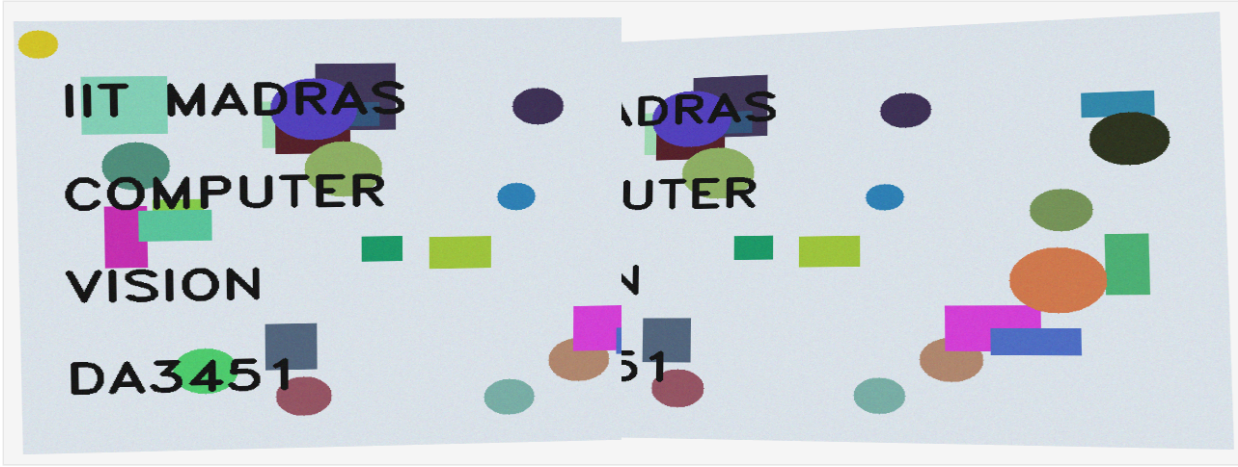
Exercise: stitch a panorama from five or six of your own photographs, using SIFT, RANSAC and your own DLT.

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Inline demonstration

image pair Poster *two views of the same planar poster — related by a homography*

1 · detect & match features | 2 · RANSAC H + stitch | two views of the same planar poster — related by a homography

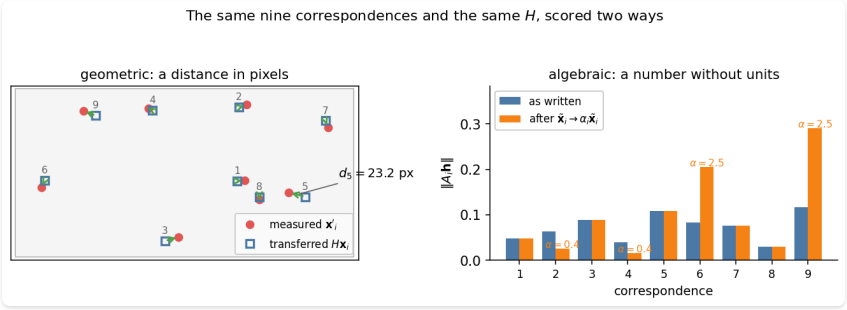


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Which error did we just minimize?

The SVD returned $\hat{\mathbf{h}} = \arg\min_{\|\mathbf{h}\|=1} \|A\mathbf{h}\|$. The minimand deserves a name. The quantity $\varepsilon_{\text{alg}} = \|A\mathbf{h}\|$ is the **algebraic error**, and it is not a distance in the image.

- **The units are meaningless.** The residual of one correspondence is the three-vector $\tilde{\mathbf{x}}'_i \times H\tilde{\mathbf{x}}_i$, whose entries mix pixels with squared pixels.
- **It depends on an arbitrary scale.** Writing $\tilde{\mathbf{x}}_i$ as $\alpha\tilde{\mathbf{x}}_i$ names the same image point, yet multiplies that point's two rows of A, and its residual, by α.
- **The constraint is arbitrary too.** Setting $\|\mathbf{h}\| = 1$ fixes the free scale of H, which had to be fixed somehow, but there is no reason for that choice to be the statistically correct one.



Left: what we want, a length in pixels. Right: what the SVD minimized, and the effect of rescaling four correspondences by α. The left picture does not move when the right one does.

The error we want has been on the RANSAC slide all along, in step 3. We have been using the geometric error to **decide**, and the algebraic error to **solve**, ever since.

$$\varepsilon_{\text{alg}} = \|A\mathbf{h}\| \qquad \text{against} \qquad \varepsilon_{\text{geom}} = \textstyle\sum_i d(\mathbf{x}'_i,\ H\mathbf{x}_i)^2$$

Three geometric errors

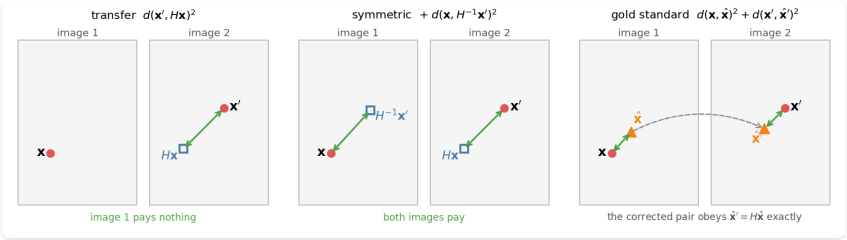
Let $d(\cdot,\cdot)$ denote Euclidean distance in the image. Each measure is a cost on H, with the measured points held fixed, and the estimate is $\hat{H} = \arg\min_H E(H)$. The three are listed in increasing fidelity, and in increasing cost.

Transfer error, charged to the second image only, which is exactly what RANSAC computed:

$$E_{\text{transfer}}(H) = \sum_i d\big(\mathbf{x}_i',\ H\mathbf{x}_i\big)^2$$

Symmetric transfer error, which repairs the asymmetry, since both measurements are noisy:

$$E_{\text{sym}}(H) = \sum_i \Big[d\big(\mathbf{x}_i',\ H\mathbf{x}_i\big)^2 + d\big(\mathbf{x}_i,\ H^{-1}\mathbf{x}_i'\big)^2\Big]$$



One correspondence, with the displacements exaggerated. Only the gold standard moves both measurements onto the model, and it pays for both moves.

Reprojection error, the gold standard. Neither measurement is trusted. Each measured pair is moved to the nearest pair $(\hat{\mathbf{x}}_i, \hat{\mathbf{x}}_i')$ that satisfies the model exactly, and the cost is the total distance moved, in both images. The corrected pairs are estimated alongside H. Under isotropic Gaussian noise this is the maximum-likelihood estimate, and also the most expensive, since the $2n$ corrected coordinates are unknowns as well.

$$E_{\text{gold}}(H) = \min_{\{\hat{\mathbf{x}}_i\}} \sum_i \Big[d(\mathbf{x}_i, \hat{\mathbf{x}}_i)^2 + d(\mathbf{x}_i', \hat{\mathbf{x}}_i')^2\Big] \quad \text{subject to} \quad \hat{\mathbf{x}}_i' = H\hat{\mathbf{x}}_i$$

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References

1. R. Hartley and A. Zisserman, *Multiple View Geometry in Computer Vision*, 2nd ed. Cambridge Univ. Press, 2004. See **§8.4** for the homography induced by a pure rotation and **§13.1** for the homography induced by a plane; then **Ch. 4, "Estimation: 2D Projective Transformations"**, in particular Alg. 4.1 for the DLT, §4.4 for normalization and §4.7 for RANSAC.
2. M. A. Fischler and R. C. Bolles, "Random Sample Consensus: A Paradigm for Model Fitting with Applications to Image Analysis and Automated Cartography," *Communications of the ACM*, 24(6):381-395, 1981.
3. R. Szeliski, "Image Alignment and Stitching: A Tutorial," *Foundations and Trends in Computer Graphics and Vision*, 2(1):1-104, 2006.

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